Eleven-dimensional Supergravity with Four Off-shell Supersymmetries

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Abstract

Off-shell supersymmetry of higher-derivative corrections to 2-derivative actions generally leads to an infinite set of ever-higher-derivative corrections. In some cases, the result is a potentially large part of an effective action for some more fundamental theory. While it is generally expected to be impossible to describe more than eight twelve? Poincaré supersymmetries this way, some constructions may be simplified considerably even if only a fraction of them closes off-shell. In this talk, I will describe eleven-dimensional supergravity with four of the 32 supersymmetries manifest and the status of a project to construct the famous $\int C \wedge X_8$ invariant.

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1 Motivation and Strategy

"M-theory" is still poorly understood. It should have (at least) 2-branes and 5-branes (not necessarily fundamental). These have comparable tensions (not parametrically-separated), so a world-volume-like description has not been found.
The alternative is to exploit the spacetime properties (*e.g.* symmetries and anomaly structure) together with known limits in the moduli space of theories (*i.e.* string theories).

We will rely on the following properties of M:

1. 11D super-Poincaré symmetry represented by the minimal multiplet as explained by Cremmer, Julia and Scherk (1978): 11D SG represented in components by a metric $g_{\boldsymbol{mn}}$, a gravitino $\psi^{\boldsymbol{\alpha}}_{\boldsymbol{m}}$, and a gauge 3-form $C_{\boldsymbol{mnp}}$.

2. Partial knowledge of the leading higher-derivative corrections to the 11D SG: $\int C \wedge X_8$, with $X_8 \sim R^4 + \cdots$ a specific contraction of the fourth exterior power of the curvature 2-form with corrections involving fewer curvatures.

3. The 4-form field strength $G = dC$ does not have integer periods on a generic spinnable Lorenzian 11-manifold. What has integral periods is instead $G - \frac{1}{4}p_1$ where $p_1 \sim \operatorname{tr} R \wedge R$ is the first Pontryagin class (Witten 1997).

4. The long-standing conjecture that this is the leading Wilsonian effective action of M-theory. Schematically,

$$\boxed{R + R^4 + 11\text{D},\ N = 1 \text{ supersymmetry} \Rightarrow \text{Leading M-theory LEEA?}} \qquad (*)$$

Now suppose there were an off-shell, 11D superspace with all 32 supercharges.
Then a description of the leading terms in the M-theory LEEA in this superspace must actually contain all orders, at least when all the auxiliary fields are integrated out.
The catch is that this superspace does not exist. (For rigid supersymmetry Berkovits (1993) explains that the algebra will not close unless the number of supercharges is 9 or less due to non-associativity of $\mathbb{O}$.)

Can we make do with fewer than 32 supercharges, say, only 4?
In this 11D, $N = 1/8$ superspace, the 4 supersymmetries would close on 4 translations giving a 4D, $N = 1$ *off-shell* subalgebra of the full 11D, $N = 1$ on-shell supersymmetry algebra.
We would then have to impose the remaining symmetries "by hand".
A good analogy was provided by Bagger and Galperin [1]: $F^2 + F^4 +$ 10D, $N = 1$ supersymmetry $\Rightarrow$ Born-Infeld.

1. We will attempt to take 4 real supersymmetries off-shell. This is the maximal amount possible with finitely many fields. These symmetries will be taken to close on 4 of the 11 translations.

2. Split the indices of the component fields of the eleven-dimensional super-Poincaré multiplet into $4 + 7$ and embed each one into supermultiplets according to their naïve 4D interpretation. The M-theory 3-form gives rise to a double complex of differential superforms (non-abelian tensor hierarchy) in this 11D, $N = 1/8$ superspace [2–4].[1]

3. Construct the 2-derivative action as a covariantization of the associated Chern-Simons invariant, and compute its field equations [4–7]. This tells us how to compare to on-shell theories and is the starting point for studying deformations.

4. Act repeatedly on the unconstrained superfields in all possible ways with super-translations and consider all possible linear combinations. This gives a large differential algebra containing connections and tensors of our reduced structure algebra. Identifying "differential ideal" is tantamount to finding the (reduced) supergeometry.

[1] Mathematically, this is a *Diff*(7)-equivariant chain complex of differential forms on Y with values in a complex of (sheaves of) representations of the 4D, $N = 1$ super-Poincaré algebra [3].

$$
\begin{tikzcd}
V \arrow[r, "D"] \arrow[d, "\bar{D}"'] & DV \arrow[r, "D"] \arrow[d, "\bar{D}"'] & D^2V \arrow[r, "D"] \arrow[d, "\bar{D}"'] & 0 \\
\bar{D}V \arrow[r, "D"] \arrow[d, "\bar{D}"'] & A_a \arrow[r, "D"] \arrow[d, "\bar{D}"'] & \bar{\boldsymbol{W}}^{\boldsymbol{\alpha}} \arrow[r, "D"] \arrow[d, "\bar{D}"'] & 0 \\
\bar{D}^2V \arrow[r, "D"] \arrow[d, "\bar{D}"'] & \boldsymbol{W}^{\boldsymbol{\alpha}} \arrow[r, "D"] \arrow[d, "\bar{D}"'] & \boldsymbol{F}_{\boldsymbol{ab}} \oplus \boldsymbol{d} \arrow[r, "D"] \arrow[d, "\bar{D}"'] & 0 \\
0 & 0 & 0 &
\end{tikzcd}
$$

$$
\begin{array}{lll}
U & DU & D^2U \\
\\
\bar{D}U & A_a & \bar{\boldsymbol{W}}^{\boldsymbol{\alpha}} \\
\\
\bar{D}^2U & \boldsymbol{W}^{\boldsymbol{\alpha}} & \boldsymbol{F}_{\boldsymbol{ab}} \quad \boldsymbol{d}
\end{array}
$$

5. Construct the full supergeometry of this 11D, $N = 1/8$ superspace so that is agrees with the 11D, $N = 1$ on-shell supergeometry in the appropriate limit. This requires finding the

   - field strength super-4-form $\boldsymbol{G}_{\boldsymbol{ABCD}}$
   - torsion 2-form $\boldsymbol{T}_{\boldsymbol{AB}}{}^{\boldsymbol{C}}$
   - curvature 2-form $\boldsymbol{R}_{\boldsymbol{AB}}{}^{\boldsymbol{cd}}$

   where $\boldsymbol{A} = \boldsymbol{a}, \alpha, \dot{\alpha}$ and $\boldsymbol{a} = a, i$.

6. Use the resulting curvature super-2-form tensor to define the super-4-form $\boldsymbol{R}^2$ and super-8-form $\boldsymbol{R}^4$, and use them to deform the 2-derivative action.

2 Putting it up to 11(D SG in $N = 1/8$ Superspace)

Following the strategy for spin-1, we first decompose 11D SG fields

$$
\begin{array}{lcll}
e_{\boldsymbol{m}}{}^{\boldsymbol{a}} & \to & e_m{}^a \ , \ e_i{}^a \ , \ e_m{}^j \ , \ e_i{}^j & \text{with} \quad a, m = 0, 1, 2, 3 \\
\psi^{\boldsymbol{\alpha}}_{\boldsymbol{m}} & \to & \psi^\alpha_m \ , \ \psi^{\alpha i}_m \ , \ \psi^\alpha_j \ , \ \psi^{\alpha i}_j \ , & \text{and} \quad i, j = 1, \ldots, 7 \\
C_{\boldsymbol{mnp}} & \to & C_{mnp} \ , \ C_{mn\,i} \ , \ C_{m\,ij} \ , \ C_{ijk} & \text{and} \quad \alpha = 1, 2
\end{array}
$$

and embed them into superfields. This requires the following surprising fact about Riemannian 7-manifolds: For any orientable and spinnable such smooth manifold, there is a 3-form φ such that

$$\sqrt{g}g_{ij} = -\tfrac{1}{144}\varepsilon^{k_1\cdots k_7}\phi_{ik_1k_2}\phi_{jk_3k_4}\phi_{k_5k_6k_7}$$

This 3-form is then embedded into a chiral superfield Φ_{ijk} together with the 35 pseudoscalars. All the other bosons get their own superfields by the naïve assignment [6]

$$\begin{aligned}
U^a &\sim \cdots + (\theta\sigma^m\bar\theta)\boldsymbol{e}_m{}^a + \theta^2(\sigma^m\bar\theta)_\alpha\boldsymbol{\psi}_m{}^\alpha + \bar\theta^2(\theta\sigma^m)_{\dot\alpha}\bar{\boldsymbol{\psi}}_m{}^{\dot\alpha} + \theta^2\bar\theta^2\boldsymbol{d}^a\\
\Psi^{\alpha i} &\sim \cdots + (\theta\sigma^m\bar\theta)\boldsymbol{\psi}_m{}^{\alpha i} + \theta^2(\bar\theta\bar\sigma_a)^\alpha\boldsymbol{y}^a_i + \bar\theta^2(\theta\sigma^{ab})^\alpha\boldsymbol{t}_{abi} + \theta^2\bar\theta^2\boldsymbol{\rho}^i_\alpha\\
\mathcal{V}^i &\sim \cdots + (\theta\sigma^m\bar\theta)\boldsymbol{e}^i_m + \bar\theta^2\theta^\alpha\eta^i_\alpha + \theta^2\bar\theta_{\dot\alpha}\bar\eta^{\dot\alpha i} + \theta^2\bar\theta^2\boldsymbol{d}^i\\
X &\sim \cdots + \theta^2 G + \bar\theta^2\bar G + \theta\sigma_m\bar\theta\,\varepsilon^{mnpq}\boldsymbol{C}_{npq} + \bar\theta^2\theta^\alpha\eta_\alpha + \theta^2\bar\theta_{\dot\alpha}\bar\eta^{\dot\alpha} + \theta^2\bar\theta^2\boldsymbol{d}_X\\
\Sigma^\alpha_i &\sim \cdots + [\theta^\alpha H_i + (\theta\sigma^{mn})^\alpha\boldsymbol{C}_{mn\,i}] + \theta^2\tilde\eta^\alpha_i\\
V_{ij} &\sim \cdots + (\theta\sigma^m\bar\theta)\boldsymbol{C}_{m\,ij} + \bar\theta^2\theta^\alpha\boldsymbol{\chi}_{\alpha\,ij} + \theta^2\bar\theta_{\dot\alpha}\bar\chi^{\dot\alpha}_{ij} + \theta^2\bar\theta^2\boldsymbol{d}_{ij}\\
\Phi_{ijk} &\sim \boldsymbol{C}_{ijk} + i\boldsymbol{F}_{ijk} + \theta^\alpha\boldsymbol{\chi}_{\alpha\,ijk} + \theta^2\boldsymbol{f}_{ijk}
\end{aligned}$$

Notes:

- Gauge fields transform canonically. Their (double) curls are field strength, torsion, and curvature 2-form components.
- Auxiliary fields are invariant.
- Compensators suffer Stückelberg shifts under "pre-gauge" transformations. They do not appear in the spectrum of gauge(-invariant) superfields.

The spins ≤ 1 fields in the list form a "non-abelian tensor hierarchy" of superfield. Such homological objects have a complex of field strengths:

$$\begin{aligned}
E_{ijkl} &= 4\partial_{[i}\Phi_{jkl]}\\
F_{ijk} &= \tfrac{1}{2i}\left(\Phi_{ijk} - \bar\Phi_{ijk}\right) - 3\partial_{[i}V_{jk]}\\
W_{\alpha ij} &= -\tfrac{1}{4}\bar{\mathcal{D}}^2\mathcal{D}_\alpha V_{ij} + 2\partial_{[i}\Sigma_{\alpha j]} + \mathcal{W}^k_\alpha\Phi_{ijk}\\
H_i &= \tfrac{1}{2i}\left(\mathcal{D}^\alpha\Sigma_{\alpha i} - \bar{\mathcal{D}}_{\dot\alpha}\bar\Sigma^{\dot\alpha}_i\right) - \partial_i X + \left[\mathcal{W}^{\alpha j}\mathcal{D}_\alpha V_{ij} + \tfrac{1}{2}\mathcal{D}^\alpha\mathcal{W}^j_\alpha V_{ij} + \text{h.c.}\,\right]\\
G &= -\tfrac{1}{4}\bar{\mathcal{D}}^2 X + \mathcal{W}^{\alpha i}\Sigma_{\alpha i}
\end{aligned}$$

It is gauged by the non-abelian KK vector multiplet $\mathcal{V}^i$ with field gauge-chiral strength $\mathcal{W}^i_\alpha$.

In addition there are some combinations of the other three superfields $\{U^a, \Psi_i, \mathcal{V}^i\}$ that appear (mixed with a little X)

$$
\begin{aligned}
S &:= \tfrac{1}{2}(G + \bar{G}) - \tfrac{1}{4}[D\sigma_a \bar{D}]U^a \\
P &:= \tfrac{1}{2i}(G - \bar{G}) + \partial_a U^a \\
X_i^a &:= \tfrac{i}{4}(\bar{D}\bar{\sigma}^a \Psi_i + D\sigma^a \bar{\Psi}_i) - \partial_i U^a \\
T_i &:= \tfrac{1}{2i}(D\Psi_i - \bar{D}\bar{\Psi}_i) + H_i
\end{aligned}
$$

2.1 "Components"

The formulæ above can be partially inverted for the (linearized) components.[2]

$$
\begin{aligned}
-4\boldsymbol{C}_{mnp} &:= \varepsilon_{mnpq}\left([D\sigma^q \bar{D}]X + 2(D^2 + \bar{D}^2)U^d\right) \\
2\boldsymbol{C}_{mn\,i} &:= (D\sigma_{mn}\Sigma_i) + (D\sigma_{mn}\Psi_i) \\
-4\boldsymbol{C}_{m\,ij} &:= [D\sigma_m \bar{D}]V_{ij} - 4\varphi_{ijk}X_a^k \\
2\boldsymbol{C}_{ijk} &:= \Phi_{ijk} + \bar{\Phi}_{ijk} + \psi_{ijkl}T^l
\end{aligned}
$$

$$
\begin{aligned}
4\boldsymbol{e}_m{}^a &:= [D\sigma_m \bar{D}]U^a - \tfrac{8}{3}\delta_m^a S \\
-4\boldsymbol{e}_m{}^i &:= [D\sigma_m \bar{D}]\mathcal{V}^i \\
4\boldsymbol{e}_i{}^b &:= \bar{D}\sigma^b \Psi_i - D\sigma^b \bar{\Psi}_i \\
4\boldsymbol{e}_i{}^j &:= \varphi^{jkl}F_{ikl} - \tfrac{1}{9}\delta_i^j \varphi^{klm}F_{klm}
\end{aligned}
$$

$$
\begin{aligned}
8i\boldsymbol{\psi}_m{}^\beta &:= \bar{D}^2 D^\beta U_m + \tfrac{4}{3}(\sigma_m \bar{D})^\beta S \\
-8i\boldsymbol{\psi}_i{}^\beta &:= \bar{D}^2 \Psi_i^\beta - \tfrac{2i}{3}\left[2(\sigma_m \bar{D})^\beta X_i^m - D^\alpha T_i - 4i\mathcal{W}_i^\alpha\right] \\
-4i\boldsymbol{\psi}_m{}^{\beta j} &:= [D\sigma_m \bar{D}]\Psi^{\beta j} - 4i\partial^j (D\bar{\sigma}_m \sigma_n)^\beta U^n - 2i(\sigma_m \bar{D})^\beta T^j - 8i(\sigma_m \bar{\boldsymbol{\psi}}^j)^\beta \\
\boldsymbol{\psi}_i{}^{\beta j} &:= ih_i{}^j(\boldsymbol{\chi}_{ijk}^\beta) + \partial_i \Psi^{\beta j} - \tfrac{i}{3}\varphi_i{}^{jk}\left[\tfrac{1}{12}\psi_k{}^{lmn}\boldsymbol{\chi}_{lmn}^\beta + iD^\beta T_k\right] - \pi_{\mathbf{14}}(\boldsymbol{\chi}_i^{\beta j})
\end{aligned}
$$

$$
\begin{aligned}
\boldsymbol{\chi}_{ij}^\alpha &:= \widehat{W}_{ij}^\alpha + i\varphi_{ijk}(\mathcal{W}^{\alpha k} - \tfrac{1}{4}\bar{D}^2 \Psi^{\alpha k}) \\
\boldsymbol{\chi}_{ijk}^\alpha &:= D^\alpha\left(F_{ijk} - 3i\varphi_{l[ij}\partial_{k]}\mathcal{V}^l\right)
\end{aligned}
$$

[2] Need to define τ, Z.

$$
\begin{aligned}
8\boldsymbol{d}^a &:= D^\beta \bar{D}^2 D_\beta U^a - \tfrac{2}{3}[D\sigma^a \bar{D}]S + 8\partial^a P \\
-4\boldsymbol{d}_{\boldsymbol{X}} &:= (D^2 + \bar{D}^2)S \\
3\boldsymbol{d}^i &:= D^\alpha \mathcal{W}^i_\alpha - \tfrac{1}{8}\left(D^\alpha \bar{D}^2 \Psi^i_\alpha + \bar{D}_{\dot\alpha} D^2 \bar{\Psi}^{\dot\alpha i}\right) + 2\partial^i P \\
8\boldsymbol{d}_{ij} &:= D^\alpha \boldsymbol{\chi}_{\alpha\, ij} + \bar{D}_{\dot\alpha} \bar{\boldsymbol{\chi}}^{\dot\alpha}_{ij} - \tfrac{8}{3}\varphi_{ijk}\partial^k S - 8\varphi_{ijk}(\tau_1)^k - 4(\tau_2)_{ij} \\
\boldsymbol{f}_{ijk} &:= -18i\bar{D}^2\left[G^{ijk,lmn}\bar{Z}_{lmn}\right] - \tfrac{1}{3}\epsilon^{ijklmnp}\partial_l \bar{Z}_{mnp} \\
2\boldsymbol{y}^a_i &:= (\bar{D}\bar{\sigma}^a \boldsymbol{\lambda}_i) \\
2\boldsymbol{t}_{abi} &:= (D\sigma_{ab}\mathcal{W}^k) + \tfrac{i}{2}\left[3(\bar{D}^{\dot\alpha}\bar{\sigma}^c\sigma_{ab}D) + (D\sigma_{ab}\sigma^c\bar{D})\right] X^k_c \\
\boldsymbol{\lambda}^i_\alpha &:= 12\boldsymbol{\psi}^i{}_\alpha + \varphi^{ijk}\boldsymbol{\chi}_{\alpha\, jk} - \tfrac{i}{6}\psi^{ijkl}\boldsymbol{\chi}_{\alpha\, jkl} \\
-4\boldsymbol{\rho}^i_\alpha &:= D^\beta \bar{D}^2 D_{(\alpha}\Psi^i_{\beta)} + 4i\partial^i\left[D_\alpha G + \tfrac{1}{2}\bar{D}^{\dot\alpha} D^2 U_{\alpha\dot\alpha}\right]
\end{aligned}
$$

Notes:

- These are all superfields.
- Supersymmetry acts by supertranslations $\delta = -\epsilon D - \bar{\epsilon}\bar{D}$.
- Gauge fields transform canonically. Their (double) curls are field strength, torsion, and curvature 2-form components. Auxiliary fields are invariant.
- Compensators suffer Stückelberg shifts under "pre-gauge" transformations. They do not appear in the spectrum of gauge(-invariant) superfields.
- There is only one dimension-1/2 invariant $\boldsymbol{\lambda}^i_\alpha$. Because of this field, the definition of the gravitino components is ambiguous. We have chosen a minimal prescription in which any such a term is absent.

2.2 Off-Shell Supersymmetry

The Wess-Zumino transformations of the 3-form field are

$$
\begin{aligned}
\delta_\epsilon \boldsymbol{C}_{abc} &= -6(\epsilon\sigma_{[ab}\boldsymbol{\psi}_{c]}) + \text{h.c.} \\
\delta_\epsilon \boldsymbol{C}_{ab\, i} &= -4i(\epsilon\sigma_{[a}\bar{\boldsymbol{\psi}}_{b]i}) - 2(\epsilon\sigma_{ab}\boldsymbol{\psi}_i) + \text{h.c.} \\
\delta_\epsilon \boldsymbol{C}_{a\, ij} &= \varphi_{ijk}(\epsilon\boldsymbol{\psi}_a{}^k) - 4i\delta_{k[i}(\epsilon\sigma_a\bar{\boldsymbol{\psi}}_{j]}{}^k) - \tfrac{1}{6}\varphi_{ijk}(\epsilon\sigma_a\bar{\boldsymbol{\lambda}}^k) + \text{h.c.} \\
\delta_\epsilon \boldsymbol{C}_{ijk} &= \varphi_{l[ij}(\epsilon\boldsymbol{\psi}_{k]}{}^l) + \text{h.c.}
\end{aligned}
$$

Note the appearance of the dimension-1/2 auxiliary field $\boldsymbol{\lambda}^i_\alpha$.

We now proceed to the off-shell metric transformations. These are unchanged from their on-shell values by our minimal choice of gravitini:

$$\begin{aligned}
\delta_\epsilon \boldsymbol{h}_{ab} &= -4(\epsilon\sigma_{(a}\bar{\boldsymbol{\psi}}_{b)}) + \text{h.c.} \\
\delta_\epsilon \boldsymbol{h}_{ai} &= -i(\epsilon\boldsymbol{\psi}_{ai}) - 2(\epsilon\sigma_a\bar{\boldsymbol{\psi}}_i) + \text{h.c.} \\
\delta_\epsilon \boldsymbol{h}_{ij} &= -2i(\epsilon\boldsymbol{\psi}_{(i,j)}) + \text{h.c.}
\end{aligned}$$

In principle there could have been a contribution by $\boldsymbol{\lambda}$ to the off-shell h_{ai} transformation, but in contrast to the 3-form transformations, it does not enter (unless we expressly redefine the gravitino components). No other contributions are possible because there is there is no other dimension-1/2 invariant with the required G_2 representations.

$$\begin{aligned}
\delta_\epsilon \boldsymbol{\psi}_a^\beta &\simeq -i(\epsilon\eta_{ab} - \epsilon\sigma_a\bar{\sigma}_b)^\beta \boldsymbol{d}^b - \tfrac{1}{6}(\bar{\sigma}_a\bar{\epsilon})^\beta \boldsymbol{d}_{\boldsymbol{X}} + \tfrac{1}{72}(\bar{\sigma}_a\bar{\epsilon})^\beta \varepsilon^{bcde}\boldsymbol{G}_{bcde} \\
\delta_\epsilon \boldsymbol{\psi}_i^\beta &\simeq -\tfrac{1}{2}(\epsilon\sigma_{ab})^\beta \omega_i{}^{ab} + \tfrac{i}{12}(\epsilon\sigma_{ab})^\beta \boldsymbol{t}_i^{ab} - \tfrac{i}{2}\epsilon^\beta \boldsymbol{d}_i + \tfrac{i}{12}(\bar{\epsilon}\bar{\sigma}_a)^\beta \boldsymbol{\Gamma}_i^a \\
\delta_\epsilon \boldsymbol{\psi}_a{}^{\beta j} &\simeq \tfrac{5i}{12}(\delta_a^b\epsilon - \epsilon\sigma_a\bar{\sigma}^b)^\beta \bar{\boldsymbol{\Gamma}}_b^j - \epsilon^\beta \left[\tilde{\omega}_a{}^{kl}\varphi_{kl}{}^j + \tfrac{i}{4}\boldsymbol{\Gamma}_a^j - \tfrac{1}{8}\left(\bar{D}\bar{\sigma}_a\boldsymbol{\lambda}^j - D\sigma_a\bar{\boldsymbol{\lambda}}^j\right)\right] \\
&\quad + (\bar{\epsilon}\bar{\sigma}_b)^\beta \left[-\delta_a^b \boldsymbol{d}_{\boldsymbol{X}}{}^j - \omega_a{}^{bj} - \tfrac{1}{2}\varepsilon_a{}^{bcd}\boldsymbol{t}_{cd}^j\right] + (\bar{\epsilon}\bar{\sigma}^{bc}\bar{\sigma}_a)^\beta \left[\tfrac{i}{3}\varphi^{jkl}\boldsymbol{G}_{bc\,kl} + \tfrac{1}{6}(\bar{D}\bar{\sigma}_{bc}\bar{\boldsymbol{\lambda}}^j)\right] \\
\delta_\epsilon \boldsymbol{\psi}_i{}^{\beta j} &\simeq -i(\epsilon\sigma^{ab})^\beta \pi_{\mathbf{14}}\boldsymbol{G}_{ab\,i}{}^j \\
&\quad - \tfrac{1}{2}\epsilon^\beta \left[2\tilde{\omega}_i{}^{kl}\varphi_{kl}{}^j + \bar{\boldsymbol{\Omega}}_i{}^j - \tfrac{1}{6}\varphi_i{}^{jk}\left(3i\boldsymbol{d}_k + \tfrac{2}{3}\varphi_k{}^{lm}\boldsymbol{d}_{lm} + \tfrac{1}{36}\psi_k{}^{lmn}\bar{\boldsymbol{f}}_{lmn}\right)\right] \\
&\quad + (\bar{\epsilon}\bar{\sigma}^a)^\beta \left[2\omega_i{}^{bj} - i\boldsymbol{h}_i^j(\boldsymbol{G}^b{}_{ijk}) + \tfrac{i}{6}\varphi_i{}^{jk}\boldsymbol{\Gamma}_k^b - \tfrac{1}{12}\varphi_i{}^{jk}\bar{D}\bar{\sigma}^b\boldsymbol{\lambda}_k\right]
\end{aligned}$$

$$\begin{aligned}
\boldsymbol{\Gamma}_i^a &:= \tfrac{i}{3!}\varepsilon^{abcd}\boldsymbol{G}_{bcd\,i} + \tfrac{1}{6}\psi_{ijkl}\boldsymbol{G}^{ajkl} + \tfrac{1}{4i}(\bar{D}\bar{\sigma}^a\boldsymbol{\lambda}_i + D\sigma^a\bar{\boldsymbol{\lambda}}_i) \\
\bar{\boldsymbol{\Omega}}_i{}^j &:= 2\partial_k[\boldsymbol{h}_{il} - 2i\partial_{(i}\mathcal{V}_{l)}]\varphi^{klj} + iD^2[\boldsymbol{h}_i{}^j - 2i\partial_{(i}\mathcal{V}^{j)}] + \tfrac{1}{6}\partial_i\psi^{jklm}Z_{klm} - q_i{}^j
\end{aligned}$$

3 Action

3.1 Chern-Simons-like Action

They also have a unique Chern-Simons-like invariant [2–4]:

$$\begin{aligned}
-12L_{CS} = i\int d^2\theta\,\Phi_3\left[\partial_1\Phi_3 G_0 + \tfrac{1}{2}W_2W_2 - \tfrac{i}{4}\bar{\mathcal{D}}^2(F_3H_1)\right] \\
+ \int d^4\theta\,V_2\left[\partial_1\left(\Phi_3 + \bar{\Phi}_3\right)H_1 + \omega(W^1, F_3) - i\mathcal{D}F_3\iota_{\mathcal{W}^1}F_3 + i\bar{\mathcal{D}}F_3\iota_{\bar{\mathcal{W}}^1}F_3\right] \\
+ i\int d^2\theta\,\Sigma_1\left[\partial_1\Phi_3W_2 - \tfrac{i}{4}\bar{\mathcal{D}}^2(F_3\mathcal{D}F_3)\right] \\
- \int d^4\theta\,X_0\left[\partial_1\left(\Phi_3 + \bar{\Phi}_3\right)F_3\right] + \text{h.c.}
\end{aligned}$$

Some observations:

- $G\Phi \wedge d\Phi$ contains the "G_2 superpotential" (Gukov 1999).
- $\frac{1}{2}\Phi \wedge W^\alpha \wedge W_\alpha$ is the holomorphic gauge-kinetic function.
- Every term is required for non-abelian tensor hierarchy gauge invariance: Had we not known this action existed, we would have discovered it by starting from any single particular term.
- There is no superpotential. The only F-term integrals of the theory are the ones that appear here.

3.2 Supervolume

The D-term integral

$$S_{vol} = -\frac{3}{\kappa^2}\int d^7y \int d^4x \int d^4\theta \, \mathrm{sdet}(E_M{}^A)(\bar{G}G)^{1/3}\sqrt{g(F)}\, f(x)$$

is uniquely fixed by [2–4]

- Local supersymmetry on 4D, $N = 1$,
- tensor hierarchy gauge invariance,
- diffeomorphism invariance in 7D,
- engineering dimensions, 4D conformal weights, $U(1)_R$ weights, and $GL(7)$ representation

$$x = \frac{H_i g^{ij}(F) H_j}{(\bar{G}G)^{2/3}}$$

Introducing the gravitino superfield Ψ_i, we can extend the local supersymmetry from 4D, $N = 1$ to all of 11D, $N = 1/8$. This fixes $f(x)$ exactly:

$$\tfrac{x}{4}\hat{f}^4 + \hat{f}^3 - 1 = 0 \quad \text{with} \quad \hat{f} := f - 2xf'$$

and the boundary condition $f(0) = 1 \Rightarrow f(x) =$ ugly but explicit function.

3.3 Components

A big mystery is how in the world this can produce the correct "scalar potential"

$$V(x, y) = -\tfrac{1}{2}R(g) + \tfrac{1}{4\cdot 4!}F^2_{ijkl} \qquad (**)$$

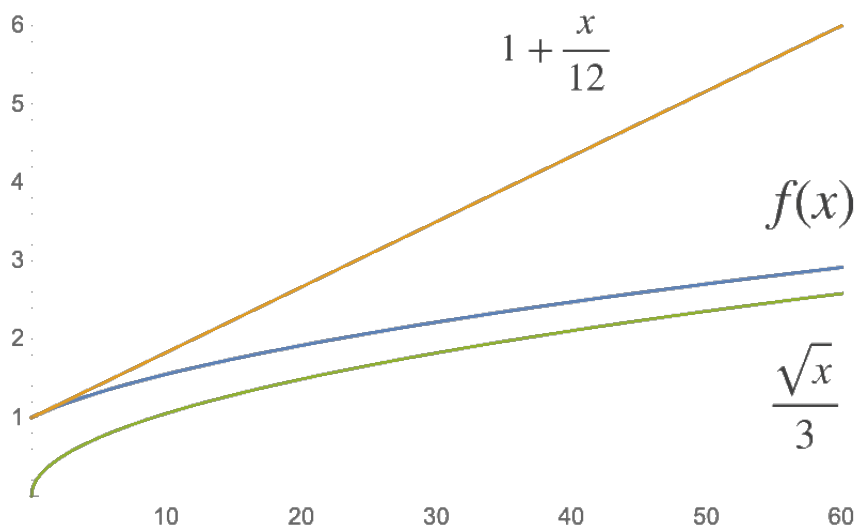


Figure 1: The function $f(x)$ appearing in the D-term integral interpolates starts as $1+\frac{x}{12}$ and slowly asymptotes to $\sqrt{x}/3$.

= Einstein-Hilbert and Maxwell terms with all polarizations along 7D.
In hybrid superspaces, this must happen by (more complicated) auxiliary field mechanism: In terms of these fields, the scalar potential is

$$V = \tfrac{1}{4}d_X^2 + 2g_{ij}\boldsymbol{d}^i\boldsymbol{d}^j - d_{\boldsymbol{7}ij}^2 + \tfrac{1}{2}d_{\boldsymbol{14}ij}^2 - \tfrac{1}{18}|f_{\boldsymbol{1}ijk} + \tfrac{i}{2}d_X F_{ijk}|^2 - \tfrac{1}{24}|f_{\boldsymbol{7}ijk}|^2 + \tfrac{1}{24}|f_{\boldsymbol{27}ijk}|^2$$

Define the "torsion forms" of φ by

$$d\varphi = \tau_0 * \varphi + 3\tau_1 \wedge \varphi + *\tau_3 \quad \text{and} \quad d(*\varphi) = 4\tau_1 \wedge *\varphi + \tau_2 \wedge \varphi$$

and define $\sigma_0, \cdots, \sigma_3$ similarly for the 3-form C_{ijk}.
Then component projection results in the equations of motion for the auxiliary fields. Plugging them in gives [5]

$$V(x,y) = -\tfrac{21}{16}\tau_0^2 - 15\tau_1^2 + \tfrac{1}{8}\tau_2^2 + \tfrac{1}{24}\tau_3^2 + \tfrac{7}{4}\sigma_0^2 + 9\sigma_1^2 + \tfrac{1}{24}\sigma_3^2 = (**)$$

3.4 Variational equations

Variation of the action with respect to prepotentials

$$\delta S = \int \left\{ \delta U^a E_{U^a} + \delta\Psi_i^\alpha E_{\Psi_i^\alpha} + \delta\mathcal{V}^i E_{\mathcal{V}^i} + \delta X E_X + \delta\Sigma_i^\alpha E_{\Sigma_i^\alpha} + \delta V_{ij} E_{V_{ij}} + \delta\Phi_{ijk} E_{\Phi_{ijk}} \right\}$$

defines the supercurrents

$$
\begin{aligned}
-9E_{U^a} &= -18\boldsymbol{d}_a + \varphi^{ijk}\boldsymbol{G}_{a\,ijk} \\
4E_{\Psi_i^\alpha} &= \boldsymbol{\lambda}_i^\alpha \\
6E_{\mathcal{V}^i} &= -6\boldsymbol{d}_i + \varphi^{jkl}\boldsymbol{G}_{ijkl} \\
96E_X &= -128\boldsymbol{d}_{\boldsymbol{X}} - i\varphi_{ijk}(\boldsymbol{f}^{ijk} - \bar{\boldsymbol{f}}^{ijk}) + \psi^{ijkl}\boldsymbol{G}_{ijkl} \\
-16E_{\Sigma_i^\alpha} &= \bar{D}^2\boldsymbol{\lambda}_\alpha^i \\
E_{V_{ij}} &= -\psi^{ijkl}\boldsymbol{d}_{kl} \\
-48E_{\Phi_{ijk}} &= \boldsymbol{f}^{ijk} - \tfrac{8i}{3}\varphi^{ijk}\boldsymbol{d}_{\boldsymbol{X}} + 18\varphi^{ijk}\varepsilon^{abcd}\boldsymbol{G}_{abcd} + 12i\varepsilon^{ijklmnp}\boldsymbol{G}_{lmnp}
\end{aligned}
$$

Notes

- These superfields satisfy differential relations expressing the pre-gauge symmetry (redundancy) in the superfield description. For example $E_\Sigma = -\frac{1}{4}\bar{D}^2 E_\Psi$ because both fields shift under a chiral spinor parameter superfield: $\delta\Psi_i = \Xi_i + D\Omega + \cdots$ and $\delta\Sigma = -\Xi + ...$ compensates.
- Going on-shell corresponds to Imposing equations of motion $E_* \to 0$.
- On-shell every auxiliary field either vanishes or becomes related to a component of the 4-form field strength $\boldsymbol{G} = dC$.

4 Supergeometry

We can now read off the low-dimension field strength, torsions, and curvatures of the 11D, $N = 1/8$ supergeometry.

- There are no dimension-0 invariants, so

$$
\boldsymbol{T}_{\alpha\beta}{}^c = 0\ ,\quad \boldsymbol{T}_{\alpha\dot\beta}{}^c = -2i(\sigma^c)_{\alpha\dot\beta}\ ,\quad \boldsymbol{T}_{\hat\alpha\hat\beta}{}^k = 0
$$

  where $\hat\alpha = \alpha, \dot\alpha$.

- At dimension 1/2, there is only the gravitino equation of motion. With our minimal choice, it only enters as the dimension-1/2 component of the 4-form

$$
\boldsymbol{G}_{\alpha bij} = \varphi_{ijk}(\sigma_b\bar{\boldsymbol{\lambda}}^k)_\alpha
$$

  This allows for the simplest set of dimension-$\frac{1}{2}$ torsions

$$
\boldsymbol{T}_{\hat\alpha\hat\beta}{}^{\boldsymbol{\gamma}} = 0\ ,\quad \boldsymbol{T}_{\alpha\boldsymbol{b}}{}^{\boldsymbol{c}} = 0
$$

  for $\boldsymbol{\gamma} = \hat\gamma, \hat\gamma k$ and $\boldsymbol{c} = c, k$

- The spin connections appearing in the supersymmetry transformations imply the dimesion-1 conventional constraints

$$\boldsymbol{T}_{\boldsymbol{ab}}{}^{\boldsymbol{c}} = 0 \tag{4.1}$$

- With this, we can proceed to solve the superspace Bianchi identities.

$$d\boldsymbol{G} + \iota_{\boldsymbol{T}}\boldsymbol{G} = 0 \quad \text{and} \quad d\boldsymbol{T} + \iota_{\boldsymbol{T}}\boldsymbol{T} = \boldsymbol{R}$$

  for the remaining field strength, torsion, and curvature components (Dragon's theorem).

For example, the dimension-1 torsions with $N = 1$ spinor index are

$$\begin{aligned} \boldsymbol{T}_{\alpha b}{}^{\gamma} &= \tfrac{i}{2}(\delta_b^c - \sigma_b\bar{\sigma}^c)_\alpha{}^\gamma \boldsymbol{d}_c \ , \quad \boldsymbol{T}_{\alpha b}{}^{\dot\gamma} = -i(\sigma_b)_\alpha{}^{\dot\gamma}\boldsymbol{R} \ , \\ \boldsymbol{T}_{\alpha j}{}^{\gamma} &= \tfrac{1}{2}\delta_\alpha^\gamma \boldsymbol{d}_{\boldsymbol{X}\,j} + \tfrac{1}{48}(\sigma^{ab})_\alpha{}^\gamma \boldsymbol{t}_{ab\,j} \ , \quad \boldsymbol{T}_{\alpha j}{}^{\dot\gamma} = -\tfrac{i}{12}(\sigma_a)_\alpha{}^{\dot\gamma}\bar{\boldsymbol{\Gamma}}^a_j \ , \end{aligned} \tag{4.2}$$

whereas the dimension-1 torsions with **7**-valued spinor index are

$$\begin{aligned} \boldsymbol{T}_{\alpha b}{}^{\gamma k} &= -\tfrac{i}{12}(\delta_b^c - \sigma_b\bar{\sigma}^c)_\alpha{}^\gamma \bar{\boldsymbol{\Gamma}}^k_c + \tfrac{1}{8}\delta_\alpha{}^\gamma \check{\boldsymbol{\Gamma}}^k_b - \tfrac{i}{32}\delta_\alpha^\gamma[\bar{D}\bar{\sigma}_b\boldsymbol{\lambda}^k + D\sigma_b\bar{\boldsymbol{\lambda}}^k] \ , \\ \boldsymbol{T}_{\alpha b}{}^{\dot\gamma k} &= i(\sigma_b)_\alpha{}^{\dot\gamma}\boldsymbol{d}_{\boldsymbol{X}}{}^k - \tfrac{1}{24}(\sigma_b)^{\underline{c}}\boldsymbol{t}_{\alpha\gamma}{}^k - \tfrac{1}{16}(\sigma_b)_{\underline{a}}\bar{\boldsymbol{t}}^{\dot\alpha\dot\gamma k} \ , \\ \boldsymbol{T}_{\alpha j}{}^{\gamma k} &\sim \tfrac{1}{2}\delta_\alpha^\gamma \bar{\boldsymbol{\Omega}}_j{}^k - \tfrac{i}{2}(\sigma^{ab})_\alpha{}^\gamma \pi_{\boldsymbol{14}}\boldsymbol{G}_{ab\,j}{}^k \ , \\ \boldsymbol{T}_{\alpha j}{}^{\dot\gamma k} &= \tfrac{i}{4}(\sigma^a)_\alpha{}^{\dot\gamma}[h_j{}^k(\boldsymbol{I}_a) - \tfrac{1}{6}\varphi_j{}^{kl}\bar{\boldsymbol{\Gamma}}^a_l] + \tfrac{i}{24}\varphi_j{}^{kl}D_\alpha\bar{\boldsymbol{\lambda}}^{\dot\gamma}_l \ . \end{aligned} \tag{4.3}$$

and so on. These results agree with our supersymmetry transformations

$$\delta E_M{}^{\boldsymbol{C}} = -\mathcal{D}_M\xi^{\boldsymbol{C}} - \xi^{\boldsymbol{D}}T_{\boldsymbol{D}M}{}^{\boldsymbol{C}} \tag{4.4}$$

5 Toward the R^2 Invariant

$$\begin{gathered} \boldsymbol{R}_{\boldsymbol{ab}}{}^{\boldsymbol{cd}} \\ \boldsymbol{T}_{\boldsymbol{ab}}{}^{\gamma}, \boldsymbol{T}_{\boldsymbol{ab}}{}^{\gamma k}, \boldsymbol{\rho}_i^\alpha \\ \boldsymbol{d}^a, \boldsymbol{d}_{\boldsymbol{X}}, \boldsymbol{d}^i, \boldsymbol{d}_{ij}, \boldsymbol{f}_{ijk}, \boldsymbol{y}_i^a, \boldsymbol{t}_k^{ab}, \boldsymbol{G}_{\boldsymbol{abcd}} \\ \boldsymbol{\lambda}_i^\alpha \end{gathered}$$

$$
\begin{aligned}
\mathbb{G} &= \boldsymbol{W}^{\alpha\beta\gamma}\boldsymbol{W}_{\alpha\beta\gamma} - \tfrac{1}{4}\bar{D}^2(\boldsymbol{d}_a^2) + \cdots \\
\mathbb{H}_i &= 4i\boldsymbol{W}_i^{\alpha\beta\gamma}\boldsymbol{W}_{\alpha\beta\gamma} - \tfrac{1}{4}D^\alpha(\bar{F}_i'^{\dot\alpha\dot\beta}\bar{D}_{\dot\beta}G_{\alpha\dot\alpha} - \tfrac{1}{8}D^2\lambda_{\alpha i}R) \\
&\quad + \tfrac{1}{24}\left(D^2\bar{D}_{\dot\alpha}\lambda_{\alpha i} + 2D^\beta\bar{D}_{\dot\alpha}D_{(\alpha}\lambda_{\beta)i}\right)G^{\alpha\dot\alpha} + \text{h.c.} + \cdots \\
\mathbb{G} &= \boldsymbol{W}^{\alpha\beta\gamma}\boldsymbol{W}_{\alpha\beta\gamma} \\
\mathbb{H}_i &= D^\gamma\boldsymbol{\mathcal{F}}_i^{\alpha\beta}\boldsymbol{W}_{\alpha\beta\gamma} - \tfrac{1}{2}\boldsymbol{\mathcal{F}}_i^{\alpha\beta}\bar{D}^{\dot\alpha}D_\beta\boldsymbol{d}_{\alpha\dot\alpha} - \tfrac{1}{2}\bar{D}^{\dot\alpha}\boldsymbol{\mathcal{F}}_i^{\alpha\beta}D_\beta\boldsymbol{d}_{\alpha\dot\alpha} \\
&\quad - \tfrac{1}{4}\left(D^\beta\bar{D}_{\dot\alpha}\boldsymbol{\mathcal{F}}_{\alpha\beta i} + \tfrac{1}{4}D_\alpha\bar{D}^2\bar{\boldsymbol{\lambda}}_{\dot\alpha i}\right)\boldsymbol{d}^{\alpha\dot\alpha} + \tfrac{1}{32}D^2\boldsymbol{\lambda}_i^\alpha D_\alpha\boldsymbol{R} + \cdots + \text{h.c.} \\
\mathbb{W}_{\alpha ij} &= \tfrac{i}{8}R_{ij}{}^{\beta\gamma}\boldsymbol{W}_{\alpha\beta\gamma} + 2i\partial_{[i}\mathbb{Z}_{\alpha j]} + i\bar{\Theta}_{[i}^{\dot\alpha\dot\beta\dot\gamma}D_\alpha\bar{\Theta}_{j]\,\dot\alpha\dot\beta\dot\gamma} - 2i\bar{R}_{ij}{}^{\dot\alpha\dot\beta}\bar{D}_{\dot\beta}G_{\alpha\dot\alpha} + \cdots \\
\mathbb{F}_{ijk} &= bR_{[ij}{}^{\alpha\beta}F'_{k]\alpha\beta} + \cdots \\
\mathbb{E}_{ijkl} &= aR_{[ij}{}^{\alpha\beta}R_{kl]\alpha\beta} + \cdots
\end{aligned}
$$

In this form, $\mathbb{H}'_y$ is manifestly real again.

Another form in which it manifestly satisfies the descent relation is

$$
\begin{array}{cccccccc}
\boldsymbol{T} & \boldsymbol{T} & & & \boldsymbol{d} & & & \\
 & \boldsymbol{T} & \boldsymbol{T} & & & \boldsymbol{d} & & \\
\boldsymbol{R} & \boldsymbol{T} & & & & & \boldsymbol{T} & \boldsymbol{T} \\
\boldsymbol{R} & \boldsymbol{\mathcal{F}} & & & & & & \\
\boldsymbol{R} & \boldsymbol{R} & & & & & &
\end{array}
$$

with

$$
\begin{aligned}
\bar{\mathbb{Z}}_i^{\dot\alpha} &:= -\tfrac{1}{2}\bar{\boldsymbol{\mathcal{F}}}_{\dot\beta\dot\gamma\,i}\bar{\boldsymbol{T}}^{\dot\alpha\dot\beta\dot\gamma} - \tfrac{1}{4}\boldsymbol{\mathcal{F}}_{\alpha\beta\,i}D^\beta\boldsymbol{d}^{\alpha\dot\alpha} + \tfrac{1}{4}D^\beta\bar{\boldsymbol{\mathcal{F}}}_i^{\dot\alpha\dot\beta}\boldsymbol{d}_{\beta\dot\beta} + \tfrac{i}{32}D^2\boldsymbol{\lambda}_{\alpha i}\boldsymbol{d}^{\alpha\dot\alpha} + \tfrac{i}{32}\bar{D}^2\bar{\boldsymbol{\lambda}}_i^{\dot\alpha}\bar{\boldsymbol{R}} + \cdots \\
\boldsymbol{\mathcal{F}}_{ab}{}^k &:= -\tfrac{4i}{3}\varphi^{ijk}\boldsymbol{G}_{ab\,jk} + \tfrac{1}{2}D\sigma_{ab}\boldsymbol{\lambda}^k
\end{aligned}
$$

$$
\bar{D}^2\mathbb{H} = 4\partial\mathbb{G}\ ,\ D\mathbb{W} - \bar{D}\bar{\mathbb{W}} = 2i\partial\mathbb{H}\ ,\ \bar{D}^2D\mathbb{F} = 4\partial\mathbb{W}\ ,\ \mathbb{E} - \bar{\mathbb{E}} = 2i\partial\mathbb{F}\ ,\ \partial\mathbb{E} = 0
$$

6 Beyond...

physical	h	ψ			ψ			h		C		C		C	χ		C	h	χ
auxiliary		d		λ	y	t	ρ		d		d_X				d			f	
compensating									η	G	η	H	η						
	6	4	12	98	84	56		21	7	1	1	21		63	21	84	63	70	140

ψ						d			R								
ψ				t			d			Γ							
ψ						Γ				d		t			Γ		λ		λ
ψ					G					G				Γ		λ		Z

	R		d_X			G			
	Γ			y	y		G		G
	Z				f			f	d

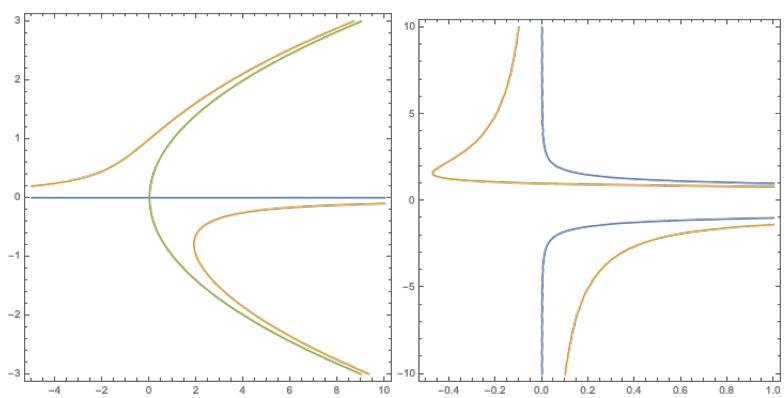
d			G		E_U					
λ		E_Ψ								
d			G		$E_\mathcal{V}$					
d_X				G		E_X		E_Φ		E_Φ
d						E_V				
f						G		G		G
						E_Φ			E_Φ	E_Φ	E_X

$$
\begin{aligned}
\boldsymbol{y}_i^a &= 2\bar{D}\bar{\sigma}^a \boldsymbol{E}_{\boldsymbol{\Psi}\,i} \\
\boldsymbol{t}_{\alpha\beta i} &= -\tfrac{1}{2}\varphi_i{}^{jk}\boldsymbol{G}_{\alpha\beta jk} - iD_{(\alpha}\boldsymbol{E}_{\boldsymbol{\Psi}\,\beta)i} \\
\boldsymbol{\rho}_i^\alpha &= -\tfrac{1}{12}(\bar{D}\bar{\sigma}^a)^\alpha\left[\psi_i{}^{jkl}\boldsymbol{G}_{a\,jkl} - i\varepsilon_a{}^{bcd}\boldsymbol{G}_{bcd\,i}\right] + 2i\psi_i{}^{jkl}\partial_j\boldsymbol{\chi}_{kl}^\alpha - i\bar{D}_{\dot\alpha}D^\alpha\overline{\boldsymbol{E}_{\boldsymbol{\Psi}}}_i^{\dot\alpha}
\end{aligned}
\tag{6.7}
$$

$$
\begin{tikzcd}
 & \boldsymbol{R} \arrow[dl, "\bar{D}"'] \arrow[dr, "D"] & & \boldsymbol{d}_a \arrow[dl, "\bar{D}\cdot"'] \arrow[dr, "\bar{D}\wedge"] & & & & \\
0 & & \boldsymbol{T}_\alpha \arrow[dl, "D"'] \arrow[dr, "\bar{D}"] & & \boldsymbol{T}_{\dot\beta\dot\alpha}{}^\gamma \arrow[dl, "\bar{D}"'] \arrow[drr, "D"] & & & \boldsymbol{T}_{\alpha\beta}{}^\gamma \arrow[dl, "D"'] \arrow[dr, "\bar{D}"] \\
 & \boldsymbol{R}_{ab}{}^{ab} & & \partial\boldsymbol{R}_{ab}{}^{ab} & & \boldsymbol{R}_{\dot\alpha\dot\beta}{}^{\gamma\delta} \oplus \partial\boldsymbol{d}^a \oplus \boldsymbol{R}_{\alpha\beta}{}^{\gamma\delta} & & 0
\end{tikzcd}
$$

$T_{\alpha b}{}^\gamma \sim \boldsymbol{d}_b$ $\quad T_{\alpha b}{}^{\dot\gamma} \sim \boldsymbol{R}$ $\quad T_{\alpha j}{}^\gamma \sim \boldsymbol{d}_j\ ,\ \boldsymbol{t}_{ab\,j}$ $\quad T_{\alpha j}{}^{\dot\gamma} \sim \bar{\boldsymbol{\Gamma}}_j^a$

$T_{\alpha b}{}^{\gamma k} \sim \boldsymbol{\Gamma}_b^k\ ,\ \bar{D}\bar{\sigma}_b\boldsymbol{\lambda}^k$ $\quad T_{\alpha b}{}^{\dot\gamma k} \sim \boldsymbol{d}^k\ ,\ \boldsymbol{t}_{ab}{}^k$ $\quad T_{\alpha j}{}^{\gamma k} \sim \boldsymbol{Z}_j{}^k\ ,\ \boldsymbol{G}_{ab\,[jk]\mathbf{14}}$ $\quad T_{\alpha j}{}^{\dot\gamma k} \sim \boldsymbol{G}_{a\,ijk}\ ,\ \boldsymbol{\Gamma}_{ai}\ ,\ D_\alpha\bar{\boldsymbol{\lambda}}_i^{\dot\gamma}$



$d \quad R \quad d \quad t \quad \Gamma \quad \lambda \quad Z \quad G \quad G$

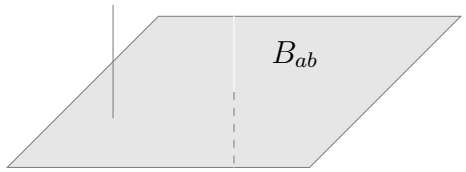


Figure 2: Properties of the Perry-Schwarz 2-form F.

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